FARMSA RESEARCH GROUP — MODULE 4

Fama–French 3-Factor Covariance

Portfolio Construction Research Note

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1. Motivation & Derivation

Why CAPM is not sufficient

A single-factor covariance model assumes all cross-asset co-movement is explained by market beta. Empirically, that is too restrictive. Size and value exposures generate persistent return comovement after controlling for the market factor.

Fama and French (1993) extend CAPM with two additional systematic factors:

Factor	Economic exposure	Construction
Mkt-RF	Broad equity risk premium	Market return minus T-bill rate
SMB	Size tilt	Small-cap minus large-cap portfolio return
HML	Value tilt	High book-to-market minus low book-to-market return

For each stock i, we estimate:

$$ r_{i,t} - r_f = \alpha_i + \beta_{i,1}(r_{m,t} - r_f) + \beta_{i,2}\,\text{SMB}_t + \beta_{i,3}\,\text{HML}_t + \varepsilon_{i,t} $$

Covariance estimator

Let B be the matrix of estimated factor loadings and Σ_f the factor covariance matrix. Under the standard assumption that residuals are cross-sectionally uncorrelated, the FF3 covariance estimator is:

$$ \hat{\Sigma}_{FF3} = B \, \Sigma_f \, B^{\top} + D $$

- $B \in \mathbb{R}^{N \times 3}$: stock-level factor loadings
- $\Sigma_f \in \mathbb{R}^{3 \times 3}$: covariance of factor returns
- $D \in \mathbb{R}^{N \times N}$ diagonal: idiosyncratic variances

For $N=50$, FF3 estimates 206 parameters versus 1,275 for the full sample covariance matrix. This reduction typically improves estimator stability out of sample.

Model risk

FF3 will understate risk when meaningful co-movement is not captured by market, size, or value factors (for example, sector-specific event risk). In those periods, residual correlations become economically relevant and the diagonal-D assumption is too strong.

In [1]:
```
import io
import urllib.request
import zipfile
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from scipy.optimize import minimize

plt.style.use("seaborn-v0_8-whitegrid")
pd.options.display.float_format = "{:,.4f}".format

DATA_DIR = Path("data")
FACTOR_URL = "https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/F-F_Resea
FACTOR_COLS = ["Mkt-RF", "SMB", "HML", "RF"]

prices = pd.read_csv(DATA_DIR / "prices.csv", index_col=0, parse_dates=True).sort_i
returns = pd.read_csv(DATA_DIR / "returns.csv", index_col=0, parse_dates=True).sort


def load_ff3_daily(url: str = FACTOR_URL) -> pd.DataFrame:
    """Download and parse Fama-French daily 3-factor data."""
    with urllib.request.urlopen(url) as response:
        payload = response.read()

    with zipfile.ZipFile(io.BytesIO(payload)) as zf:
        csv_name = [name for name in zf.namelist() if name.lower().endswith(".csv")
        text = zf.read(csv_name).decode("utf-8")

    lines = text.splitlines()
    start = next(
        i
        for i, line in enumerate(lines)
        if line.strip()[:1].isdigit() and len(line.split(",")[0].strip()) == 8
    )

    records = []
    for line in lines[start:]:
        row = line.strip()
        if not row or not row[0].isdigit():
            break
        records.append(row)
```

```
    ff = pd.read_csv(
        io.StringIO("\n".join(records)),
        header=None,
        names=["date", *FACTOR_COLS],
    )
    ff["date"] = pd.to_datetime(ff["date"], format="%Y%m%d")
    ff = ff.set_index("date").sort_index() / 100.0
    return ff


ff = load_ff3_daily()
common_dates = returns.index.intersection(ff.index)
returns = returns.loc[common_dates]
ff = ff.loc[common_dates, FACTOR_COLS]

print(f"Assets: {returns.shape[1]} | Observations: {returns.shape[0]}")
print(f"Sample window: {returns.index.min().date()} to {returns.index.max().date()}


def ff3_covariance(returns_window: pd.DataFrame, ff_window: pd.DataFrame) -> np.nda
    """FF3 covariance: B * Cov(factors) * B' + diag(idiosyncratic variance)."""
    factors = ff_window[["Mkt-RF", "SMB", "HML"]].to_numpy()
    risk_free = ff_window["RF"].to_numpy()
    excess_returns = returns_window.to_numpy() - risk_free[:, None]

    betas = np.linalg.lstsq(factors, excess_returns, rcond=None)[0].T
    residuals = excess_returns - factors @ betas.T

    factor_cov = np.cov(factors, rowvar=False, ddof=1)
    idiosyncratic_var = residuals.var(axis=0, ddof=1)

    return betas @ factor_cov @ betas.T + np.diag(idiosyncratic_var)


test_cov = ff3_covariance(returns.iloc[:252], ff.iloc[:252])
n_assets = returns.shape[1]

assert test_cov.shape == (n_assets, n_assets)
assert np.allclose(test_cov, test_cov.T)
assert np.min(np.linalg.eigvalsh(test_cov)) > -1e-8

print("FF3 covariance estimator sanity check passed.")
```

```
Assets: 50 | Observations: 1759
Sample window: 2018-01-03 to 2024-12-30
FF3 covariance estimator sanity check passed.
```

In [2]:

```
def portfolio_variance(weights: np.ndarray, cov_matrix: np.ndarray) -> float:
    return float(weights @ cov_matrix @ weights)


def min_variance_weights(cov_matrix: np.ndarray) -> np.ndarray:
    n = cov_matrix.shape[0]
    w0 = np.full(n, 1.0 / n)

    # Tiny diagonal bump improves optimizer stability without changing economics.
```

```
    cov_matrix = cov_matrix + np.eye(n) * 1e-10

    result = minimize(
        portfolio_variance,
        w0,
        args=(cov_matrix,),
        method="SLSQP",
        bounds=[(0.0, 1.0)] * n,
        constraints=[{"type": "eq", "fun": lambda w: w.sum() - 1.0}],
        options={"ftol": 1e-12, "maxiter": 1000},
    )

    return result.x if result.success else w0


LOOKBACK_DAYS = 252
REBALANCE_EVERY = 21
START_AFTER = 252

strategy_names = ["Equal Weight", "Sample Cov MinVar", "FF3 Cov MinVar"]
strategy_values = {name: [1.0] for name in strategy_names}
current_weights = {name: np.full(returns.shape[1], 1.0 / returns.shape[1]) for name

rebalance_dates = []
weights_history = {name: [] for name in strategy_names}

backtest_dates = returns.index[START_AFTER:]

for t, current_date in enumerate(backtest_dates):
    idx = returns.index.get_loc(current_date)

    if t % REBALANCE_EVERY == 0:
        window_returns = returns.iloc[idx - LOOKBACK_DAYS: idx]
        window_ff = ff.iloc[idx - LOOKBACK_DAYS: idx]

        current_weights["Equal Weight"] = np.full(returns.shape[1], 1.0 / returns.s
        current_weights["Sample Cov MinVar"] = min_variance_weights(window_returns.
        current_weights["FF3 Cov MinVar"] = min_variance_weights(ff3_covariance(win

        rebalance_dates.append(current_date)
        for name in strategy_names:
            weights_history[name].append(current_weights[name].copy())

    day_returns = returns.iloc[idx].to_numpy()
    for name in strategy_names:
        strategy_values[name].append(strategy_values[name][-1] * (1.0 + current_wei

portfolio_value = pd.DataFrame(
    strategy_values,
    index=[backtest_dates[0] - pd.Timedelta(days=1), *backtest_dates],
)

weights_frames = {
    name: pd.DataFrame(weights_history[name], index=rebalance_dates, columns=return
    for name in strategy_names
}
```

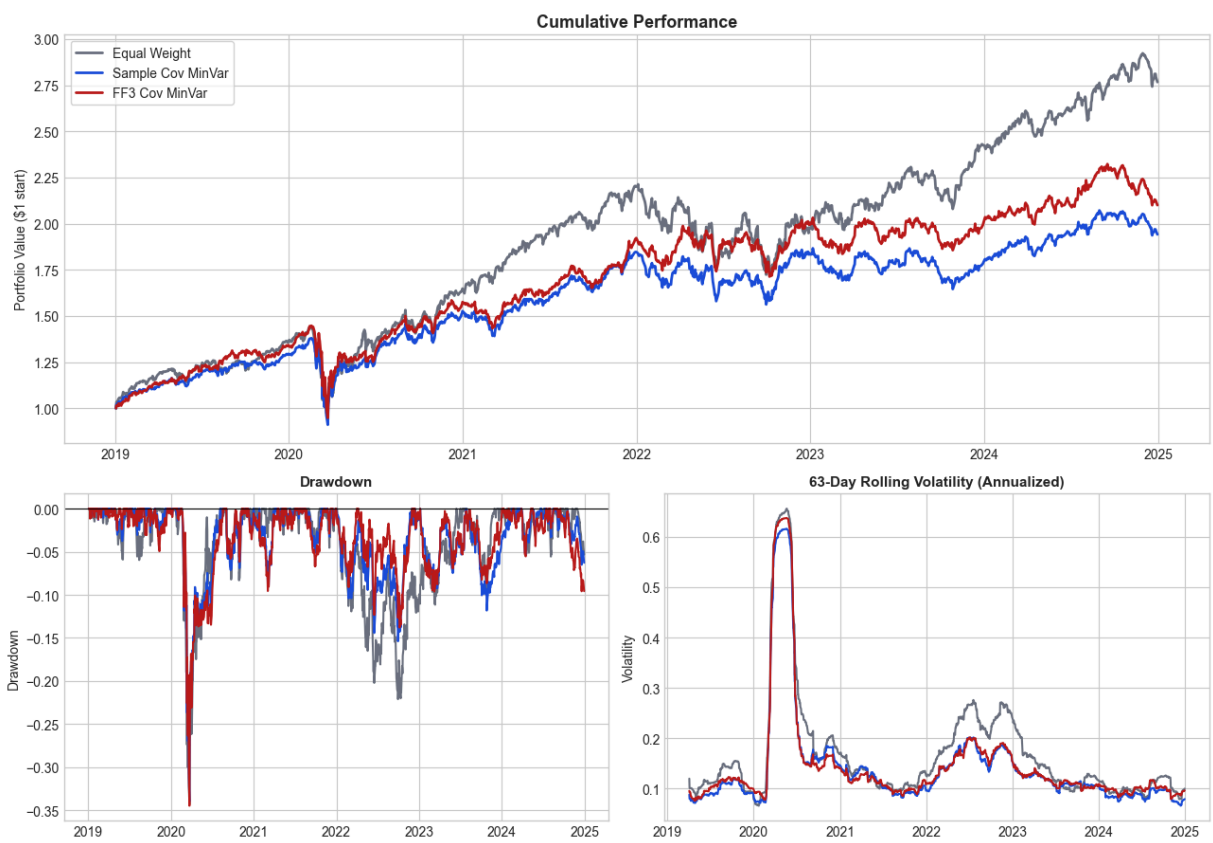
```
print("Backtest complete.")
```

```
Backtest complete.
```

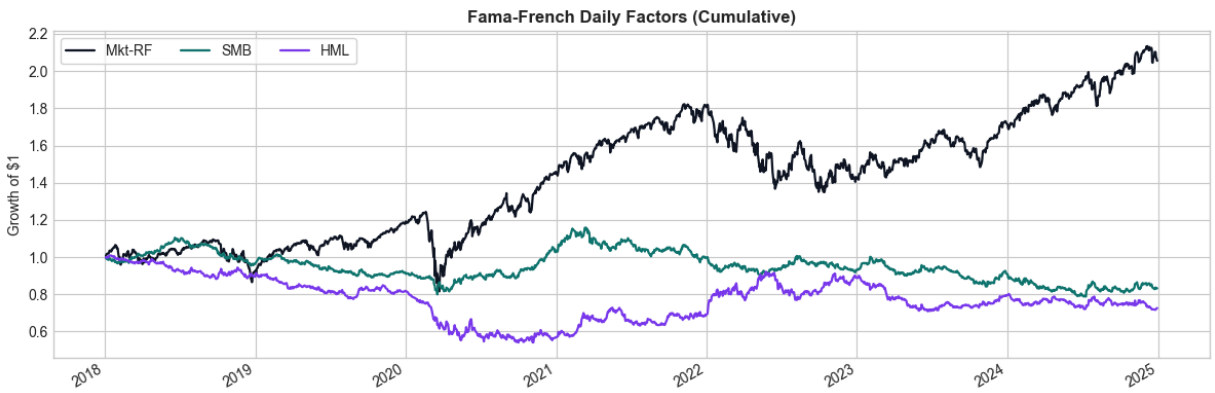
In [3]:

```
colors = {
    "Equal Weight": "#6b7280",
    "Sample Cov MinVar": "#1d4ed8",
    "FF3 Cov MinVar": "#b91c1c",
}

portfolio_returns = portfolio_value.pct_change().dropna()
drawdown = portfolio_value.div(portfolio_value.cummax()).sub(1.0)
rolling_vol = portfolio_returns.rolling(63).std() * np.sqrt(252)

fig = plt.figure(figsize=(13, 9))
grid = fig.add_gridspec(2, 2, height_ratios=[1.25, 1.0])

ax_perf = fig.add_subplot(grid[0, :])
for name in portfolio_value.columns:
    ax_perf.plot(portfolio_value.index, portfolio_value[name], label=name, color=co
ax_perf.set_title("Cumulative Performance", fontsize=13, fontweight="bold")
ax_perf.set_ylabel("Portfolio Value ($1 start)")
ax_perf.legend(loc="upper left", frameon=True)

ax_dd = fig.add_subplot(grid[1, 0])
for name in drawdown.columns:
    ax_dd.plot(drawdown.index, drawdown[name], color=colors[name], linewidth=1.5)
ax_dd.axhline(0.0, color="black", linewidth=0.8)
ax_dd.set_title("Drawdown", fontsize=11, fontweight="bold")
ax_dd.set_ylabel("Drawdown")

ax_vol = fig.add_subplot(grid[1, 1])
for name in rolling_vol.columns:
    ax_vol.plot(rolling_vol.index, rolling_vol[name], color=colors[name], linewidth
ax_vol.set_title("63-Day Rolling Volatility (Annualized)", fontsize=11, fontweight=
ax_vol.set_ylabel("Volatility")

plt.tight_layout()
plt.show()


def annualized_turnover(weights_df: pd.DataFrame, rebalance_every: int) -> float:
    if len(weights_df) <= 1:
        return 0.0
    one_way = weights_df.diff().abs().sum(axis=1).dropna()
    return float(one_way.mean() * (252 / rebalance_every))


def summary_stats(series: pd.Series) -> pd.Series:
    daily = series.pct_change().dropna()
    ann_return = (1.0 + daily.mean()) ** 252 - 1.0
    ann_vol = daily.std() * np.sqrt(252)
    sharpe = ann_return / ann_vol if ann_vol > 0 else np.nan
    max_dd = drawdown[series.name].min()
```

```
    return pd.Series(
        {
            "Ann.Return %": ann_return * 100,
            "Ann.Vol %": ann_vol * 100,
            "Sharpe": sharpe,
            "MaxDD %": max_dd * 100,
        }
    )


stats = pd.DataFrame(
    {name: summary_stats(portfolio_value[name]) for name in portfolio_value.columns
).T
stats["Turnover %"] = [
    annualized_turnover(weights_frames[name], REBALANCE_EVERY) * 100
    for name in stats.index
]
stats = stats.round(2).sort_values("Sharpe", ascending=False)

display(stats)

factor_cumulative = (1.0 + ff[["Mkt-RF", "SMB", "HML"]]).cumprod()

fig, ax = plt.subplots(figsize=(12, 4))
factor_cumulative.plot(ax=ax, linewidth=1.6, color=["#111827", "#0f766e", "#7c3aed"
ax.set_title("Fama-French Daily Factors (Cumulative)", fontsize=12, fontweight="bol
ax.set_ylabel("Growth of $1")
ax.legend(loc="upper left", ncol=3, frameon=True)
plt.tight_layout()
plt.show()
```



	Ann.Return %	Ann.Vol %	Sharpe	MaxDD %	Turnover %
Equal Weight	20.9200	19.8600	1.0500	-34.2700	0.0000
FF3 Cov MinVar	14.9900	17.6600	0.8500	-34.4600	226.6800
Sample Cov MinVar	13.4000	17.1800	0.7800	-33.9700	404.1400



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5. Interpretation

Use the chart and summary table to evaluate FF3 on risk-adjusted outcomes, not just headline return.

1. **FF3 vs sample covariance**: Compare realized volatility and Sharpe side by side. If FF3 delivers similar return with lower realized vol, the factor structure is adding useful regularization.

2. **FF3 vs CAPM (M3)**: The relevant question is whether SMB/HML exposures add incremental explanatory power after market beta. Outperformance should show up through more stable risk forecasts and better ex-post efficiency.

3. **Stress behavior**: Review drawdown windows (for example 2020 and 2022). During market stress, correlation regimes shift and idiosyncratic terms can become correlated; this is where static factor models are most fragile.

4. **Implementation fit**: FF3 is generally appropriate for diversified multi-sector equity universes with dispersion in size and value profiles. It is less informative in narrow universes where names share similar factor loadings.

Module 4 complete — refer to the integration notebook for a cross-model comparison.